| | | |
|---|---|---|
| | | $a = -6.0 \times 10 \text{ m s}$ |
| **3** | **B** | Considering car and trailer as a single system to find common acceleration<br>$F_{net} = F_D - f_{car} - f_{trailer} = 5600 - 1200 - 400 = 4000 \text{ N}$<br>$a = \dfrac{4000}{1500 + 500} = 2.0 \text{ m s}^{-2}$<br>Considering only the trailer:<br>$F_{net} = m_{trailer} a = 500(2.0) = 1000 \text{ N}$<br>$F_{net} = T - f_{trailer}$<br>$1000 = T - 400$<br>$T = 1400 \text{ N}$ |
| 4 | C | Taking moments at P, force at Q must be upwards to balance the larger clockwise |